

Space Marching for the Steady Quasi-1D Euler Equations

We will now examine the possibility of space marching the steady quasi-1D Euler equations. We will look at the non-conservative form and fit the shock, if it exists. In conservative form, the steady quasi-1D Euler equations may be written as:

$$\frac{\partial \rho u A}{\partial x} = 0 \quad \text{Continuity Equation}$$

$$\frac{\partial (\rho u^2 + p) A}{\partial x} = p \frac{\partial A}{\partial x} \quad \text{X - Momentum Equation}$$

$$\frac{\partial (e + p) u A}{\partial x} = 0 \quad \text{Energy Equation}$$

In nonconservative form this becomes:

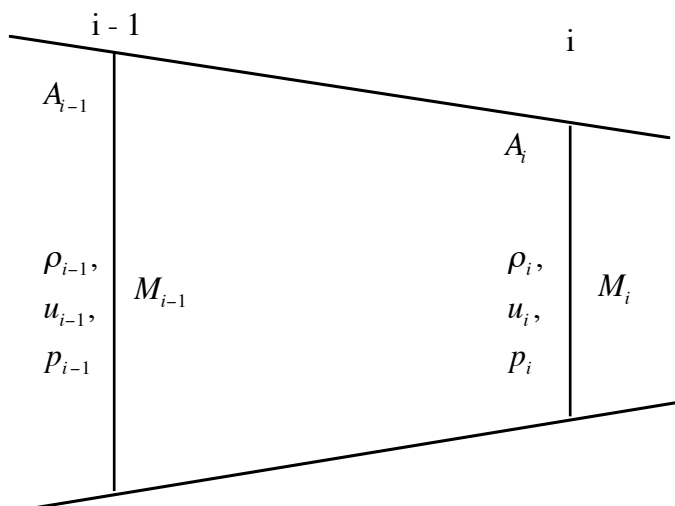
$$u A \frac{\partial \rho}{\partial x} + \rho A \frac{\partial u}{\partial x} + \rho u \frac{\partial A}{\partial x} = 0 \quad \text{Continuity Equation}$$

$$u^2 A \frac{\partial \rho}{\partial x} + 2 \rho u A \frac{\partial u}{\partial x} + A \frac{\partial p}{\partial x} + (\rho u^2 + p) \frac{\partial A}{\partial x} = p \frac{\partial A}{\partial x} \quad \text{X - Momentum Equation}$$

$$\frac{1}{2} u^3 A \frac{\partial \rho}{\partial x} + \left[\frac{\gamma}{\gamma - 1} p A + \frac{3}{2} \rho u^2 A \right] \frac{\partial u}{\partial x} + \frac{\gamma}{\gamma - 1} u A \frac{\partial p}{\partial x} = - \left[\frac{\gamma}{\gamma - 1} p u + \frac{3}{2} \rho u^3 \right] \frac{\partial A}{\partial x} \quad \text{Energy Equation}$$

In matrix form the differential form can be written as:

$$\begin{bmatrix} u & \rho & 0 \\ u^2 & 2\rho u & 1 \\ \frac{1}{2}u^3 & \frac{\gamma}{\gamma-1}p + \frac{3}{2}\rho u^2 & \frac{\gamma}{\gamma-1}u \end{bmatrix} \begin{pmatrix} \delta \rho \\ \delta u \\ \delta p \end{pmatrix} = \begin{pmatrix} -\rho u \frac{\delta A}{A} \\ -\rho u^2 \frac{\delta A}{A} \\ -\left[\frac{\gamma}{\gamma-1} p u + \frac{1}{2} \rho u^3 \right] \frac{\delta A}{A} \end{pmatrix}$$



Change in Flow Due to Area Change

After a great deal of algebraic manipulation this leads to:

$$\delta\rho = \rho \frac{M^2}{1 - M^2} \frac{\delta A}{A}$$

We can rewrite this as:

$$\rho_i = \rho_{i-1} + \rho \frac{M^2}{1 - M^2} \frac{A_i - A_{i-1}}{A} \quad (1)$$

Now, from continuity we have:

$$\rho u A = \text{constant}$$

or

$$u_i = \frac{\rho_{i-1} u_{i-1} A_{i-1}}{\rho_i A_i} \quad (2)$$

And from energy we have:

$$(e + p)uA = \text{constant}$$

Therefore,

$$\frac{e + p}{\rho} = h_i = h + \frac{1}{2}u^2 = \frac{\gamma}{\gamma - 1} \frac{p}{\rho} + \frac{1}{2}u^2 = \text{constant}$$

or

$$p_i = \rho_i \left[\frac{\gamma - 1}{2\gamma} (u_{i-1}^2 - u_i^2) + \frac{p_{i-1}}{\rho_{i-1}} \right] \quad (3)$$

For an initial guess we can freeze the coefficient in Eq. 1 at the old value, upstream:

$$\rho \frac{M^2}{1 - M^2} \frac{1}{A} = \rho_{i-1} \frac{M_{i-1}^2}{1 - M_{i-1}^2} \frac{1}{A_{i-1}} \Rightarrow \quad (4)$$

$$\bar{\rho}_i, \quad \bar{u}_i, \quad \bar{p}_i \Rightarrow \bar{M}_i$$

This would make the solution first order accurate in space. For a better guess (second order accurate), we can iterate using:

$$\rho \frac{M^2}{1 - M^2} \frac{1}{A} = \left(\frac{\bar{\rho}_i + \rho_{i-1}}{2} \right) \frac{\left(\frac{\bar{M}_i + M_{i-1}}{2} \right)^2}{1 - \left(\frac{\bar{M}_i + M_{i-1}}{2} \right)^2} \left(\frac{2}{\bar{A}_i + A_{i-1}} \right) \quad (5)$$

But, what about the possibility of a shock that may exist in the quasi-1D nozzle? **We can use shock fitting!**

$$A_1 = A_2$$

$$\begin{array}{c|c} \rho_1, & \rho_2, \\ u_1, & u_2, \\ p_1 & p_2 \\ M_1 & M_2 \end{array}$$

Jump Across a Shock

At a shock, we know that the area is constant (the shock jump occurs over a very short distance). Thus, we can use the equation:

$$\rho_2 = \rho_1 \left(1 + \frac{M_1^2 - 1}{\frac{\gamma - 1}{2} M_1^2 + 1} \right) \quad (6)$$

Then, once again continuity and energy results in:

$$u_2 = \frac{\rho_1 u_1}{\rho_2} \quad (7)$$

and

$$p_2 = \rho_2 \left[\frac{\gamma - 1}{2\gamma} (u_1^2 - u_2^2) + \frac{p_1}{\rho_1} \right] \quad (8)$$

After, the shock the solution can be marched once again!

If in this problem, the shock location was not known, but the back pressure at the downstream exit was specified then we would need to iterate for the shock location that gives the correct back pressure!

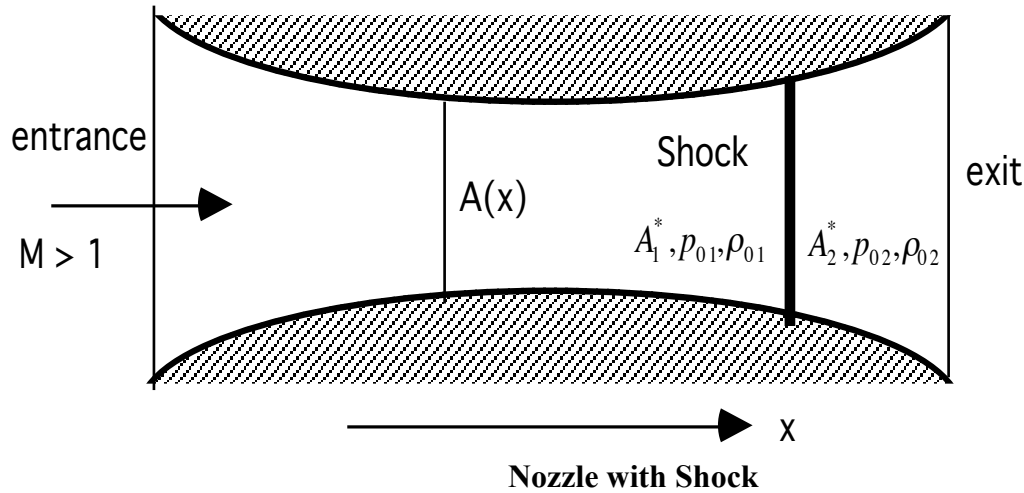
Why might we want to use this as an initial solution for a CFD problem?

- (1) This solution will be conservative, since the shock is exactly fitted.
- (2) The shock is in the correct position, although not smeared.
- (3) The solution will contain some of the same errors in space as the full CFD.
- (4) The smearing of the shock will result in small transients, not large transients that may result in rapid shock motion and instabilities.

We can compare this to the exact solution for flow in a nozzle with a possible shock. At any point in the nozzle we can compare the current dependent variables to their stagnation quantities:

$$1 + \frac{\gamma - 1}{2} M^2 = \frac{T_0}{T} = \left(\frac{p_0}{p} \right)^{\frac{\gamma - 1}{\gamma}} = \left(\frac{\rho_0}{\rho} \right)^{\gamma - 1} = \left(\frac{a_0}{a} \right)^2$$

and remember that the stagnation quantities remain unchanged in regions of isentropic flow.



(1) Given the inlet Mach number one can directly determine the upstream sonic area:

$$A^* = AM \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (9)$$

as well as the stagnation pressure and density.

(2) At any point prior to the shock one must iterate for the Mach number using

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (10)$$

being careful to pick the correct branch (supersonic). Once the local Mach number is determined the pressure, density and velocities can be calculated based on stagnation quantities.

(3) At the shock location the new Mach number immediately downstream of the shock is given in terms of the Mach number immediately upstream:

$$M_2^2 = \frac{1 + \frac{\gamma-1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}} \quad (11)$$

Then using the local area at the shock and equation (9) the new sonic area is obtained. The density and pressure can also be given as functions of the upstream Mach number:

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma+1)M_1^2}{2 + (\gamma-1)M_1^2}; \quad \frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \quad (12)$$

and new stagnation quantities can be calculated.

(4) Finally, downstream of the shock the flow is isentropic and equation (10) can be iterated using the new sonic area being careful to pick the correct branch (subsonic), along with the stagnation relationships.